WHAT PLAYS THE ROLE OF A CONIC: THE LAB'S CONSTRUCTIONS AND OVALS, OVOIDS AND POLARITIES

DEGREE–DIAMETER LAB

Abstract. A sequel to `SYNTHETIC.md` (branch `claude/synthetic-incidence-geometry-s4m08f`), written for Gabriela Araujo-Pardo and Ignacio López, who asked what happens with conics in generalized polygons. We first say what a conic is combinatorially (an oval, and the set of absolute points of a polarity) and what plays these two roles in generalized quadrangles and hexagons. Then we look at each construction of the lab through these objects. In a projective plane of odd order the answer is sharp: if the frame of the three-hat or four-hat construction is placed relative to a conic (the special point is the pole of the special line, two marked points are the points of contact of the tangents from it, and the other two marked points are conjugate), the polarity of the conic extends to a polarity of the whole structure (Theorem 1.4). This explains the links and the rotation, gives an explicit regular completion (the polarity itself), and folds the structure into a graph of diameter 2 with q^2+q+6 vertices and maximum degree $q+2$. A conic-adapted choice does not give a fifth hat: every such frame is infeasible in the hat model for $q \le 13$, and no self-polar five-hat extension exists at $q = 5$. In quadrangles and hexagons the conic-like objects (ovoids, partial ovoids, spreads, elliptic quadric sections) are already the ingredients of Theorems A–D and of the diameter-7 freed sets; we restate how, give Theorem A as a statement about pairs of conics, and record what the polarity picture gives there. No lab value is improved. Every statement is labelled [new], [restatement] or [standard], and [proof], [computer] or [heuristic].

0. Conventions and the dictionary

We use the conventions of `SYNTHETIC.md` §0: an incidence structure has points, lines and a symmetric incidence I; its incidence graph has the elements as vertices and the flags as edges; distances and the diameter are taken there. For the hat constructions we use the frame and the six rules (R1)–(R6) of `SYNTHETIC.md`, Definition 1.1, with the same names: anti-flag (A, ℓ), marked points X_s on ℓ ($s \in S$), lines $\lambda_s = AX_s$, a distinguished label a, $T = S \setminus \{a\}$, links π and rotation ρ (permutations of T), new points Q, H_s and new lines M, K_s.

0.1. **What a conic is, combinatorially.** In $\mathrm{PG}(2,q)$ a *conic* $\mathcal{C}$ is a set of $q+1$ points, no three collinear (an *oval*); for q odd every oval is a conic (Segre). For q odd a conic is also the set of absolute points $(p \,\mathrm{I}\, p^{\perp})$ of an *orthogonal polarity* $\perp$: an incidence-preserving involution exchanging points and lines. Both descriptions have analogues in generalized polygons, and they are different objects there:

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	projective plane	generalized quadrangle	generalized hexagon
"no two too close" (arc / partial ovoid)	arc: no 3 collinear; oval: $q+1$ points ($q+2$, hyperoval, q even)	partial ovoid: pairwise non-collinear; ovoid: $st+1$ points, one on every line	distance-2 (partial) ovoid: non-collinear (q^4+q^2+1 points, one on every line; known only for $q=2,3,4$); distance-3 (partial) ovoid: pairwise opposite (q^3+1 points)
absolute points of a polarity	conic (q odd); a line (q even, pseudo-polarity); Hermitian curve (q square)	Suzuki–Tits ovoid of $W(q)$, $q=2^{2e+1}$ (the only case with a polarity); absolute lines form a spread	Ree–Tits ovoid of $H(q)$, $q=3^{2e+1}$ (the only case with a polarity); absolute lines form a spread
dual object	dual conic (tangent lines)	spread	spread (Hermitian spread, every q)
quadric sections	conic = plane section of a quadric	$Q(4,q)$: elliptic sections $Q^-(3,q)$ are ovoids; plane sections are conics ($q+1$ pairwise non-collinear points)	$Q(6,q)$: elliptic sections $Q^-(3,q)$ are partial ovoids (q^2+1); the lines in an elliptic hyperplane $Q^-(5,q)$ form a spread (Hermitian spread)

[standard] [7, 4, 6, 2]. The lab's constructions use the first and third rows throughout (Sections 2 and 3). The second row, the polarity, is what explains the hats (Section 1).

0.2. **Polarities fold incidence structures.** A *polarity* of an incidence structure $\mathcal{S}$ is a bijection θ from points to lines and lines to points with $\theta^2 = \mathrm{id}$ and $p \mathrel{\mathrm{I}} L \iff \theta(L) \mathrel{\mathrm{I}} \theta(p)$. A point p is *absolute* if $p \mathrel{\mathrm{I}} \theta(p)$. The *polarity graph* G_θ has the points of $\mathcal{S}$ as vertices, $p \sim r$ iff $p \neq r$ and $p \mathrel{\mathrm{I}} \theta(r)$ (symmetric because θ is a polarity). For $\mathrm{PG}(2,q)$ with an orthogonal polarity this is the Erdős–Rényi graph.

Lemma 0.1 (Folding)**.** [standard; proof] *Let $\mathcal{S}$ have a polarity θ, all degrees at most Δ and diameter $D \geq 2$. Then G_θ has maximum degree at most Δ and diameter at most $D-1$, and $|V(G_\theta)|$ is half the number of elements of $\mathcal{S}$.*

Proof. The degree of p in G_θ is the number of lines through p, minus one if p is absolute. Map each element x of a path in the incidence graph to x if it is a point and to $\theta(x)$ if it is a line. Consecutive elements $u \mathrel{\mathrm{I}} L$ go to u and $\theta(L)$, which are equal or adjacent in G_θ (as $u \mathrel{\mathrm{I}} \theta(\theta(L))$). So a path of length m from p to an element y gives a walk of length at most m from p to y or $\theta(y)$. For points $p \neq r$, $d(p,r)$ is even and $d(p,\theta(r))$ is odd, both at most D; the smaller of the two is at most $D-1$. □

So a self-polar bipartite graph of diameter D gives a graph of diameter $D-1$ with half the vertices. In this note it is used in two ways: as a description of what the conic does to the hats (§1.3), and to compare with the tables of the general degree–diameter problem (where it gives nothing new; §§1.3, 2.4, 3.2).

0.3. **Cloning, and why the cloned sets are ovals and ovoids.** A *clone* of a point x is a new point incident with exactly the lines through x; a *line clone* of L is a new line incident with exactly the points of L (`SYNTHETIC.md` §2.1 calls the second a shadow line with all stand-ins original).

Lemma 0.2 (Cloning)**.** [standard; proof] *Let $\mathcal{S}$ have diameter $D \geq 2$, and add a clone x' for each entry x of a multiset X of points. Then distances between old elements do not change, $d(x',y) = d(x,y)$ for every element $y \neq x$, and $d(x',x) = 2$; so the diameter is still D. A line L gains $|L \cap X|$ incidences (with multiplicity) and no old point gains any. Dually for line clones; and point clones and line clones may be added together (a clone of x and a clone of L are at distance at most 3 if $x \mathrel{\mathrm{I}} L$, and at the distance of x and L otherwise, plus the flag $x' \mathrel{\mathrm{I}} L'$ if one adds it).*

Proof. A shortest path from x to $y \neq x$ leaves x along a line through x, which also passes through x'; conversely a path from x' is a path from x after its first step. Two clones of one point have a common line. □

So adding points never hurts the diameter; it only costs degree. If every old element may gain k incidences, the point multisets that can be cloned are exactly those with at most k entries on every line. This is where the conic-like objects come from [restatement]:

- *projective plane,* $k = 1$: any two points are collinear, so only one point can be cloned. The hats of §1 are clones of the marked points of one line ℓ that *miss* ℓ (each H_s is on the q lines through X_s other than ℓ); everything else in the construction repairs what missing ℓ breaks;
- *projective plane,* $k = 2$: an arc can be cloned; an oval (a conic) and, dually, its tangent lines give Delorme's construction at $\Delta = q + 3$ (§1.6);
- *generalized quadrangle,* $k = 1$: partial ovoids; an ovoid gives a regular result (Theorem B with weights, `SYNTHETIC.md` §2; §2 below);
- *generalized hexagon,* $k = 1$: non-collinear sets (partial distance-2 ovoids). A distance-2 ovoid with a distance-2 spread would give $2(q+2)(q^4+q^2+1) = 2N + 2(q^4+q^2+1)$ elements at $\Delta = q + 2$, fewer than Theorem D's $2N + 2q^4 + 4q^3 + 4q + 10$, which doubles along pairwise opposite elements with levels instead.

In the plane the conic therefore enters in two different ways: as a set that can be cloned when the degree may rise by two, and (Section 1) as the absolute set of the polarity that organizes the hats when it may rise by one.

1. Projective planes: the hats relative to a conic

1.1. **The conic frame.** Let q be odd, $\mathcal{C}$ a conic of $\Pi = \mathrm{PG}(2, q)$ and $\perp$ its polarity. A point off $\mathcal{C}$ is *external* (on two tangents) or *internal* (on none). We use the following standard facts [standard].

Proposition 1.1 (Conjugation on a polar line)**.** *Let $A \notin \mathcal{C}$ and $\ell = A^\perp$. For a point X of ℓ, the polar line $X^\perp$ passes through A; write $X^\tau = X^\perp \cap \ell$. Then τ is an involution of the points of ℓ and $X^\perp = AX^\tau$. Its fixed points are the points of $\ell \cap \mathcal{C}$: two points Z, Z' if A is external (then AZ, AZ' are the two tangents from A, touching $\mathcal{C}$ at Z and Z'), none if A is internal. If A is external, X^τ is the harmonic conjugate of X with respect to Z, Z'; X and X^τ are* conjugate *points.*

So there are two matchings between the points of ℓ and the lines through A: the frame matching $X \mapsto AX$, which rule (R1) uses, and the polar matching $X \mapsto X^\perp = AX^\tau$. They differ by τ. The marked points are now chosen as follows.

Definition 1.2 (Conic frame)**.** Let A be external, $\ell = A^\perp$, $\ell \cap \mathcal{C} = \{Z, Z'\}$, and $X \in \ell \setminus \mathcal{C}$. *Three hats:* $(X_a, X_0, X_1) = (Z, X, X^\tau)$. *Four hats:* $(X_a, X_c, X_0, X_1) = (Z, Z', X, X^\tau)$. In words: A is the pole of ℓ; the marked points are the two points of contact of the tangents from A (only one of them for the three hats) and a pair of points of ℓ that separates them harmonically. The lines $\lambda_a = AZ$ and $\lambda_c = AZ'$ are the tangents from A, and $\lambda_0 = (X^\tau)^\perp$, $\lambda_1 = X^\perp$.

Figure 1 shows the frame. For every odd $q \geq 3$ such frames exist ($\ell \setminus \mathcal{C}$ has $q - 1 \geq 2$ points). The cross-ratio $(X_a, X_c; X_0, X_1)$ of a four-hat conic frame is -1; conversely every four-hat frame with $(X_a, X_c; X_0, X_1) = -1$ is a conic frame for each conic of the pencil of conics tangent to AX_a at X_a and to AX_c at X_c. [standard]

1.2. **The polarity extends.** Let τ also denote the permutation of labels with $X_{\tau(s)} = X_s^\tau$; in a conic frame τ swaps 0 and 1 and fixes a (and c). The next lemma is stated for any polarity θ of a projective plane with $\theta(A) = \ell$; then τ, defined by $X^\tau = \theta(X) \cap \ell$, is again an involution of ℓ whose fixed points are the absolute points of ℓ.

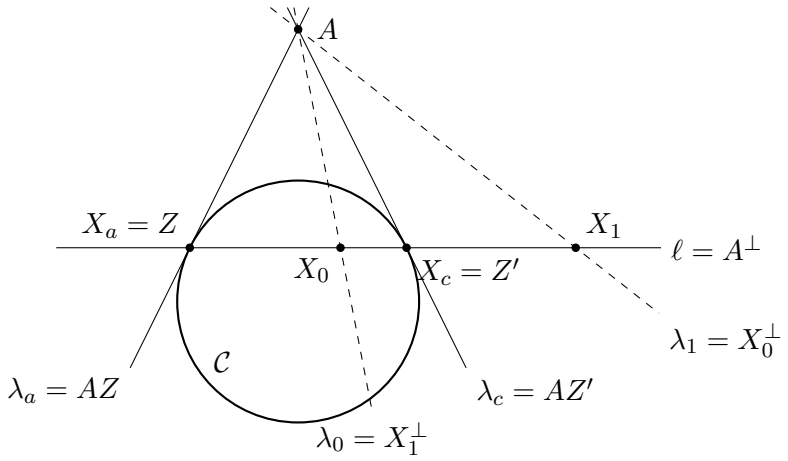


FIGURE 1. The conic frame of the four hats (drawn over the reals). A is external, ℓ is its polar, the chord of contact ZZ'; the marked points are Z, Z' and a pair X_0, X_1 that separates them harmonically, so that AX_0 is the polar of X_1 and AX_1 the polar of X_0. The three hats use Z, X_0, X_1.

Lemma 1.3. [new; proof] *Let Π be a projective plane with a polarity θ, A a non-absolute point, $\ell = \theta(A)$, and let the marked set $\{X_s\}$ be τ-invariant. Suppose*

$$\tau(a) = a, \qquad \tau\pi\tau = \pi^{-1}, \qquad \tau\rho\tau = \rho^{-1} \quad \textit{(on } T\textit{)}.$$

Then $\hat{\theta}$, equal to θ on Π and with $H_s \leftrightarrow K_{\tau(s)}$, $Q \leftrightarrow M$, is a polarity of the structure $\mathcal{S} = \mathcal{S}(\Pi; A, \ell, (X_s), \pi, \rho)$ of `SYNTHETIC.md`, Definition 1.1.

Proof. $\hat{\theta}$ is an involution exchanging points and lines; we show that it maps each flag (p, L) of $\mathcal{S}$ to a flag $(\hat{\theta}(L), \hat{\theta}(p))$. Note $\theta(X_s) = AX_{\tau(s)} = \lambda_{\tau(s)}$ and hence $\theta(\lambda_s) = X_{\tau(s)}$.

(R1) θ maps flags of Π to flags of Π. The deleted flags (X_s, λ_s) go to $(\theta(\lambda_s), \theta(X_s)) = (X_{\tau(s)}, \lambda_{\tau(s)})$, again deleted. An added flag $(X_t, \lambda_{\rho(t)})$ goes to $(X_{\tau\rho(t)}, \lambda_{\tau(t)})$; with $u = \tau\rho(t)$ we have $\rho(u) = \rho\tau\rho(t) = \tau(t)$, because $\tau\rho\tau = \rho^{-1}$ gives $\rho\tau\rho = \tau$. So it is the added flag $(X_u, \lambda_{\rho(u)})$.

(R2)↔(R3) If N I X_s, $N \neq \ell$, then $\theta(N)$ is a point of $\theta(X_s) = \lambda_{\tau(s)}$ other than $\theta(\ell) = A$, so $(\theta(N), K_{\tau(s)})$ is a flag by (R3). The converse is the same computation.

(R4)↔(R5) A point $p \in \ell$ goes to the line $\theta(p) = Ap^\tau$ through A. If p is unmarked, so is p^τ, and $\theta(p) \in \Lambda^*$; $\theta(X_a) = \lambda_a \in \Lambda^*$; and $\theta(X_t) = \lambda_{\tau(t)}$ with $\tau(t) \in T$. So θ maps ℓ^* onto Λ^*, and $(p, M) \mapsto (Q, \theta(p))$ preserves (R4)/(R5). $(H_s, M) \mapsto (Q, K_{\tau(s)})$.

(R6) $(A, K_a) \mapsto (H_a, \ell)$. A link $(H_t, K_{\pi(t)})$ goes to $(H_{\tau\pi(t)}, K_{\tau(t)})$, and with $u = \tau\pi(t)$, $\pi(u) = \pi\tau\pi(t) = \tau(t)$; so it is the link of u. □

Theorem 1.4 (Self-polar hats)**.** [new; proof and computer] *Let q be odd. In a conic frame (Definition 1.2) the polarity of the conic, extended by*

$$H_a \leftrightarrow K_a, \quad H_c \leftrightarrow K_c, \quad H_0 \leftrightarrow K_1, \quad H_1 \leftrightarrow K_0, \quad Q \leftrightarrow M,$$

is a polarity of the three-hat structure of [1] *and of the four-hat structure.*

Proof. Lemma 1.3 with $\tau = (0\ 1)$. Three hats: $\pi = (0\ 1)$ commutes with τ and is an involution, $\rho = \mathrm{id}$. Four hats: $\pi = \rho = \sigma = (0\ c\ 1)$ and $\tau\sigma\tau = (1\ c\ 0) = \sigma^{-1}$. □

Computer. `conics/polar_frames.py` checks the theorem for every conic frame (every external A, every choice of Z, X) for $q = 3, 5, 7$, and for random frames at $q = 9, 11, 13$; four independent squad programs (`squads/conics-r1/c1`–`c4`) re-derived it from the rules for $q = 3, \ldots, 13$ and over GF(9), GF(25) (163 instances), and showed for $q = 5, 7$ that, with θ and τ as above, *exactly* the

conic frames are self-polar (2,496 labelled frames tried). The same lemma gives self-polar three hats for q even: the absolute points of a pseudo-polarity form a line m, so $\ell = \theta(A)$ has exactly one absolute point X_a and $\tau = (0\ 1)$ as before, unless $\ell = m$, in which case every point of ℓ is absolute, τ is the identity and any three marked points will do. For q square a unitary polarity has $\sqrt{q}+1$ absolute points on ℓ, which gives self-polar three and four hats. Checked for $q = 4, 8, 16$ (pseudo) and $q = 4, 9, 16$ (unitary), and independently by squad card `c8` on every non-absolute A at $q = 4, 8$ (pseudo) and $q = 4, 9$ (unitary): 6,830 instances, all self-polar with the right τ (the card's two "failures" are the two $\ell = m$ cases, where its brief wrongly assumed $\tau \neq \mathrm{id}$).

Remark 1.5 (Why the conic frame is forced). [proof] For the four hats, a permutation τ of S as in Lemma 1.3 must fix a, preserve T and invert the 3-cycle σ; so it is a transposition on T and fixes exactly two labels, a and one $t \in T$. Hence X_a and X_t are absolute points of ℓ: for an orthogonal polarity, the two points of contact of the tangents from A. For the three hats, τ fixes a and either swaps $0, 1$ or fixes them; the second needs three absolute points on ℓ, impossible for a conic. So among frames with A the pole of ℓ, the conic frames are the only ones to which the lemma applies; the squad check above confirms that no other labelled frame is self-polar with this θ.

Without fixing θ the same holds, by a group computation [computer]: squad card `conics-r2/c6` computed with `nauty` the full automorphism group of the incidence graph (collineations and correlations) of the four-hat structure for every cross-ratio $t = (X_a, X_c; X_0, X_1)$ at $q = 5, 7$. A polarity exists exactly for $t \in \{-1, 2, \frac{1}{2}\}$, i.e. exactly when X_a and one point X_u, $u \in T$, separate the other two harmonically (a conic frame up to renaming the labels of T, which the 3-cycle σ permits); at $q = 7$ the two equianharmonic frames ($t = 3, 5$) have no correlation at all and 72 collineations, against 24 for the harmonic ones. All harmonic four-hat structures are isomorphic (120 and 336 labelled harmonic frames at $q = 5, 7$, one canonical form), as they must be: the stabilizer of (A, ℓ) induces $\mathrm{PGL}(2, q)$ on ℓ, which is transitive on harmonic quadruples.

1.3. **What the gadget is, in polar terms.** Under the polarity every line is θ of a point, so every incidence $p \mathrel{\mathrm{I}} L$ is a symmetric relation "p is conjugate to $\theta(L)$" between two points. Reading Table 3 of `SYNTHETIC.md` this way:

Proposition 1.6. [new; proof] *In a four-hat conic frame:*

- *(a) the four deleted flags $(X_s, \lambda_s) = (X_s, X_{\tau(s)}^{\perp})$ are exactly the conjugate pairs among the marked points: Z and Z' with themselves (they are absolute), and $\{X_0, X_1\}$;*
- *(b) the three rotation flags $(X_0, \lambda_c), (X_c, \lambda_1), (X_1, \lambda_0)$ make X_0 and Z' conjugate (two flags, $X_0 \mathrel{\mathrm{I}} Z'^{\perp}$ and $Z' \mathrel{\mathrm{I}} X_0^{\perp}$) and X_1 absolute ($X_1 \mathrel{\mathrm{I}} X_1^{\perp}$);*
- *(c) the links are the conjugacies H_0–H_c (two flags) and H_1 absolute, and $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$ are the two flags of the conjugacy A–H_a;*
- *(d) Q is conjugate to every H_s, to Z and to the unmarked points of ℓ; H_s is conjugate to the q points of $\lambda_{\tau(s)}$ other than A.*

So the surgery on the old points replaces the conjugacy relation on the marked set $\{Z, Z', X_0, X_1\}$ (two absolute points and one conjugate pair) by another one (one conjugate pair, X_0–Z', and one absolute point, X_1), and the links do the same on the hats.

Proof. (a) $\lambda_s = AX_s = \theta(X_{\tau(s)})$. (b) $\lambda_c = AZ' = Z'^{\perp}$, $\lambda_1 = X_0^{\perp}$, $\lambda_0 = X_1^{\perp}$. (c) $K_c = \hat{\theta}(H_c)$, $K_1 = \hat{\theta}(H_0)$, $K_0 = \hat{\theta}(H_1)$, $K_a = \hat{\theta}(H_a)$, $\ell = \theta(A)$. (d) $K_s = \hat{\theta}(H_{\tau(s)})$; $M = \hat{\theta}(Q)$ contains $\ell^* = \{Z\} \cup \{\text{unmarked}\}$; a line $N \neq \ell$ through X_s is θ of a point of $\theta(X_s) = \lambda_{\tau(s)}$ other than A. □

For the three hats in a conic frame the only deleted flag is $(Z, Z^{\perp})$ (an absolute point loses its loop), and the links $H_0 \mathrel{\mathrm{I}} K_1$, $H_1 \mathrel{\mathrm{I}} K_0$ make H_0 and H_1 absolute.

Corollary 1.7 (Folding the hats)**.** [new; proof and computer] *For every odd prime power $q \geq 3$ the polarity graph of the four-hat conic frame is a graph of diameter 2 and maximum degree $q+2$ on q^2+q+6 vertices; that of the three hats has q^2+q+5 vertices.*

Proof. Theorem 1.4, Lemma 0.1 and Theorem 1.7 of `SYNTHETIC.md` (diameter 3, degrees at most $q+2$). □

By Proposition 1.6, this graph is the Erdős–Rényi graph of $\mathrm{PG}(2,q)$ with the edge X_0X_1 replaced by X_0Z', plus five vertices: Q joined to Z, to the $q-3$ unmarked points of ℓ and to the four hats; H_s joined to the q points of $\lambda_{\tau(s)}$ other than A; and H_a–A, H_0–H_c. (We checked by computer that this description is exactly the polarity graph, $q = 3,\dots,13$: `conics/folded_direct.py`.) *Comparison* [computer]: at $q = 13$ this is 188 vertices at $(\Delta, D) = (15, 2)$ for general graphs. The table of the general problem lists 187 (E. Canale, 2012) in the UPC table and 190 (2026) in the current Wikipedia table, read on 29 October 2026. So the folded four hats lie between the two and are not a record; at $\Delta = 11, 13$ ($q = 9, 11$: 96 and 138) they are far below the tabled 106 and 162. The three-hat fold at $q = 13$ has 187 vertices, the old tabled value; we do not know how Canale's graph was built.

The four hats can also be built without changing any incidence of the plane (`research/plus3/plus3.tex`, Proposition "the four hats without changing the plane"): labels $a, 1, 2, 3$, a 3-cycle $s = (1\ 2\ 3)$, and the rotation carried by the hats (H_i also on $AX_{s(i)}$, K_i also through $X_{s^{-1}(i)}$). Its conic frame is $X_a = Z$, $X_2 = Z'$, $\{X_1, X_3\}$ conjugate, with $\tau = (1\ 3)$, which inverts s.

Proposition 1.8 (Self-polar four hats on an unchanged plane)**.** [new; proof and computer] *In this conic frame the zero-change four-hat structure is self-polar, with $H_x \leftrightarrow K_{\tau(x)}$ and $Q \leftrightarrow M$. Its polarity graph is the Erdős–Rényi graph of* $\mathrm{PG}(2,q)$*, with* no edge changed*, plus five vertices: a graph of diameter* 2 *and maximum degree $q+2$ on q^2+q+6 vertices.*

Proof. The same check as in Lemma 1.3, rule by rule over the table of the cited proposition; the only new pairs are the extra incidences, and $(H_i, AX_{s(i)})$ goes to $(X_{\tau s(i)}, K_{\tau(i)})$, which is the extra incidence of $K_{\tau(i)}$ because $s^{-1}\tau = \tau s$. No incidence of the plane is changed, so the old part of the polarity graph is the Erdős–Rényi graph. □

Computer. `conics/zero_polar.py`, $q = 3,\dots,13$ (polarity, diameter 3, old part of the fold equal to the Erdős–Rényi graph, fold of diameter 2); independently squad card `conics-r2/c1`, 42 instances ($q = 3,\dots,13$, every conjugate pair), with $\tau = \mathrm{id}$ failing as a control.

So, in the language of the general problem: *the Erdős–Rényi graph of* $\mathrm{PG}(2,q)$*, q odd, has room for five more vertices at maximum degree $q+2$ and diameter* 2, and the five vertices are the folded hats. We have not searched the literature on extensions of polarity graphs; the count q^2+q+6 is below the tabled values except at $\Delta = 15$, as said above.

1.4. **An explicit regular version.** In the four-hat structure the elements of degree $q+1$ (the deficient ones) are the points of Π off ℓ, other than A, on the lines AY with Y unmarked, and dually (`SYNTHETIC.md`, Lemma 1.4). `SYNTHETIC.md` completes the structure by any perfect matching of non-incident deficient pairs (König). The conic gives one.

Proposition 1.9 (The polarity completes the hats)**.** [new; proof and computer] *In a conic frame (three or four hats), let p be a deficient point. If $p \notin \mathcal{C}$ put $\mu(p) = p^\perp$; if $p \in \mathcal{C}$, let p^* be the second point of $\mathcal{C}$ on the line Ap and put $\mu(p) = (p^*)^\perp$, the tangent at p^*. Then μ is a bijection from the deficient points onto the deficient lines with $p \not\mid \mu(p)$, so adding the flags $(p, \mu(p))$ makes the structure $(q+2)$-regular (still of diameter* 3*). The completed structure is again self-polar.*

Proof. Let p be deficient, on AY with $Y \in \ell$ unmarked. $p^\perp$ does not pass through A (as $p \notin \ell$), is not ℓ (as $p \neq A$), and meets $\ell = A^\perp$ in the pole of the line $pA = AY$. By Proposition 1.1, $AY = (Y^\tau)^\perp$, so this pole is Y^τ. The marked set is τ-invariant, so Y^τ is unmarked and $p^\perp$ is deficient. If $p \in \mathcal{C}$, the line AY is not a tangent (the tangents from A are AZ, AZ' and Z, Z' are marked in the four hats; in the three hats Z' is unmarked, but AZ' meets $\mathcal{C}$ only in $Z' \in \ell$, which

is not deficient), so AY is a secant and $p^* \neq p$ is deficient too, $p^* \notin \ell$; its tangent $(p^*)^\perp$ meets ℓ in Y^τ as before and does not contain p. The map $p \mapsto p^\perp$ is injective, and on $\mathcal{C}$ we composed it with the involution $p \mapsto p^*$ of the deficient points of $\mathcal{C}$, so μ is injective; the numbers of deficient points and lines are equal. Self-polarity: $\hat{\theta}$ maps the added flag $(p, p^\perp)$ to itself and $(p, (p^*)^\perp)$ to $(p^*, p^\perp)$, the added flag of p^*. □

Computer. `conics/polar_frames.py` for $q = 3, \dots, 13$, and independently squad card `c3` (24 instances, $q = 3, \dots, 13$; the number of deficient points on $\mathcal{C}$ varies with X, from 0 to 12 at $q = 13$). The same matching also completes the zero-change four hats of Proposition 1.8 (their deficient elements are again the points off ℓ, other than A, on the lines AY, Y unmarked, and dually; checked for $q = 3, \dots, 13$ in `conics/zero_polar.py`): this gives a $(q+2)$-regular structure of diameter 3 that contains $\mathrm{PG}(2, q)$ with every incidence kept, is self-polar, and has every flag given by an explicit rule. In the folded graph the added flags $(p, p^\perp)$ are loops, so the regular bipartite version does not give a regular graph of diameter 2.

1.5. **Does a conic allow more hats?** *The question in polar terms.* By Corollary 1.7 a self-polar extension of $\mathrm{PG}(2, q)$ by k new points and k new lines, of maximum degree $q + 2$ and diameter 3, is the same as an extension of the Erdős–Rényi graph by k vertices of diameter 2 in which any two points have a common line; the four hats are $k = 5$, and five hats would be $k = 6$.

Heuristic: what the conic cannot change [heuristic]. For large q, any extension by a few new elements is a *copy configuration* (`research/plus3/plus3.tex`, Theorem 3.4): every new point copies an old point and every new line an old line, and the line joining two copied points must itself be copied or carry a deleted flag. A conic offers a second natural set of points to copy, an arc. But k copied points on a conic have $\binom{k}{2}$ joining lines (secants), all of which must be copied or broken; on a line the join is one line (ℓ, copied by M). So collinear marked points are forced, and the conic can only choose *which* points of ℓ are marked and how the gadget is wired. Theorem 1.4 says that the conic chooses them so that the structure is self-polar; it does not create room for a fifth hat.

Computer [computer]:

(i) *Hat model at conic frames* (`conics/sat_hatmodel.py`, the model of `research/plus2/sat_gadget.py`: bulk incidences fixed, every incidence of the gadget free, degree $\leq q + 2$, diameter 3). Five hats at Z and two conjugate pairs, and six hats at Z, Z' and two conjugate pairs, one Q and one M: UNSAT for every choice of the pairs, $q = 5, 7, 9, 11, 13$, with and without imposing self-polarity (35 five-hat and 35 six-hat frames, 140 runs, each under 4 s; `conics/hatmodel_conic.out`). The four-hat conic frame is SAT in all these runs (control).

(ii) *All self-polar extensions at $q = 5$* (`conics/sat_polar.py`): new points and new lines tied by the conic polarity, all incidences of the new elements free, no change to the plane. $k = 5$: SAT. $k = 6$ (five hats): UNSAT. The search is split soundly by the orbit of the conic group $\mathrm{PGL}(2, 5)$ that contains the point most often copied (conic point, external, internal) and the number r of lines through it used by one new point; see `conics/k6_q5_E0.out` for the cases and times.

Earlier work, which this note does not repeat, ruled out five hats in the hat model at default and random frames for $q = 4, \dots, 13$, and +3 without changes for $q \geq 42$ (`research/plus2/REPORT.md`, `research/plus3/plus3.tex`).

1.6. **At $\Delta = q + 3$ the conic itself is cloned.** [restatement] Let q be odd, $\mathcal{C}$ a conic. Clone every point of $\mathcal{C}$ and every tangent line of $\mathcal{C}$, and add the flag (clone of p, clone of the tangent at p) for each $p \in \mathcal{C}$. By Lemma 0.2 the diameter stays 3. A secant gains two clones, a tangent one, a passant none; dually an external point is on two tangent clones, a point of $\mathcal{C}$ on one, an internal point on none; the clones have degree $q + 2$. So the maximum degree is $q + 3$ and there are $2(q^2 + q + 1) + 2(q + 1) = 2(q^2 + 2q + 2)$ elements: 394 at $q = 13$, the 2013 table value at

$(16, 3)$ (Delorme's construction; `research/d3d6/REPORT.md`). The lab's 396 there is not of this kind: it places three four-hat gadgets on three anti-flags permuted by a Singer subgroup of order 3 (`research/d3d6/REPORT.md`). Could self-polar gadgets for *one* conic do better? At $\Delta = q + 3$ every element may gain two incidences. A gadget in a conic frame $F = \{A, Z, Z', X, X^\tau\}$ raises by one exactly the lines meeting F ($5q-2$ of them) and, by the polarity, the points whose polars meet F; so several such gadgets fit only if no line meets three of their frames. [computer] At $q = 13$ no three frames have this property (`conics/gadgets16.py`, exhaustive over the 546 conic frames up to the conic group), so at most two conic-adapted gadgets fit, and the conic does not reach the value 396.

Remark 1.10. [new; proof] No hat gadget can be added on top of the conic clones at $\Delta = q + 3$: a gadget raises by one the degree of every point of its line ℓ (`SYNTHETIC.md`, Lemma 1.4: each point of ℓ gains M, K_t or the net $+1$ of the rotation), every line of $\mathrm{PG}(2, q)$ contains external points ($q \geq 3$), and external points already have degree $q + 3$. The structure is self-polar (clone of $p \leftrightarrow$ clone of the tangent at p), and its fold has $q^2 + 2q + 2$ vertices at $(\Delta, D) = (q+3, 2)$: 197 at $(16, 2)$, below the tabled 200 and 216.

Computer. `conics/conic_clones.py`, $q = 3, \ldots, 13$: $2(q^2 + 2q + 2)$ elements, maximum degree $q + 3$, diameter 3, a point of degree $q + 3$ on every line, self-polar, fold of diameter 2.

2. Generalized quadrangles

2.1. **The analogues.** In a generalized quadrangle of order (s, t):

- an *ovoid* is a set of $st + 1$ pairwise non-collinear points meeting every line once, a *partial ovoid* a set of pairwise non-collinear points, an *m-ovoid* a set meeting every line in m points, and a *weighted m-ovoid* a multiset doing so; dually spreads and covers [standard];
- in $Q(4, q)$ the elliptic quadric sections $Q^-(3, q)$ are ovoids for every q (and every ovoid of $Q(4, p)$, p prime, is one [2]); a plane section of $Q(4, q)$ that is a conic is a set of $q+1$ pairwise non-collinear points, the closest analogue of a conic *inside* the quadrangle; in $W(q)$, q odd, there is no ovoid, and the elliptic quadrics of $\mathrm{PG}(3, q)$ (ovoids of the space, not of $W(q)$) are what the lab uses [standard];
- $W(q)$ has a polarity if and only if $q = 2^{2e+1}$; its absolute points form the Suzuki–Tits ovoid and its absolute lines a spread [7] [standard]. This is the analogue of "conic = absolute points". For q odd there is none: $W(q)$ and $Q(4, q)$ are dual and not isomorphic.

2.2. **Where they appear in Theorems A–C.** [restatement] Theorem B (`SYNTHETIC.md`, Theorem 2.2) adds clones over a weight w on the points and shadow lines over a multiset $\mathcal{M}$ of lines; for a regular result w must be a weighted $(\Delta - s - 1)$-ovoid and $\mathcal{M}$ a weighted $(\Delta - t - 1)$-cover. So the ovoid-like object is exactly the place where clones go, and the spread-like object where shadow lines go:

cells	quadrangle	clones on	shadow lines over	conic-like content
$(q + 1 + j, 4)$, q even	$W(q)$	$j\times$ an ovoid	$j\times$ a spread	ovoid (elliptic quadric or Tits)
Theorem A: $(11, 4), (13, 4), (15, 4)$	$\mathrm{AS}(q)$	$\mathcal{E}_1 \cup \mathcal{E}_2$ (weighted 2-ovoid)	none	two elliptic quadrics; pairs of conics (Prop. 2.1)
Prop. C: $(16, 4)$	$W(13)$	a 2-ovoid	$2\times$ a spread	union of two elliptic quadrics (Ebert's partition)
$(7, 4)$: 359	$Q(4, 5)$	an ovoid ($Q^-(3, 5)$)	21 lines, a partial spread	ovoid; partial spread at Tallini's bound

The limits are also conic-like statements: for q odd, $W(q)$ has no ovoid and $Q(4, q)$ no spread, which is why $\mathrm{AS}(q)$ and weights are used at $\Delta = q + 2$; on $Q(4, q)$ with one ovoid of clones the

shadow lines form a partial spread (or nearly), so at most q^2-q+2 of them (Tallini's bound q^2-q+1 for partial ovoids of $W(q)$, `POLISHED.md` Remark 3.5), and the lab's searches found $52, 72, 92$ at $q = 9, 11, 13$ against the $61, 97, 141$ needed to beat Theorem A (`research/gq4x/REPORT.md`). Large partial ovoids of $W(3, q)$ in the literature [3] are far smaller than this.

2.3. **Theorem A as a statement about pairs of conics.** Theorem A (`POLISHED.md` §4) works in $\mathrm{PG}(3, q)$, q odd, with the symplectic form $\beta(u, v) = u_0v_1-u_1v_0+u_2v_3-u_3v_2$, the point $x = (1{:}0{:}0{:}0)$, a non-square ε and the elliptic quadrics $\mathcal{E}_1 : 2X_0X_1 = X_2^2 - \varepsilon X_3^2$ and $\mathcal{E}_2 : 2\varepsilon X_0X_1 = X_2^2 - \varepsilon X_3^2$, both through x with tangent plane $x^\perp$. The lines of $\mathrm{AS}(q)$ not through x are the totally isotropic lines $L \not\ni x$ minus their point z in $x^\perp$, and L lies in the plane $z^\perp$, which contains x.

Proposition 2.1. [restatement; proof and computer] *For every point $z \neq x$ of $x^\perp$, the plane $z^\perp$ meets $\mathcal{E}_1$ and $\mathcal{E}_2$ in two conics $\mathcal{C}_1(z), \mathcal{C}_2(z)$ through x with the same tangent xz at x, and from the point z of this common tangent every other line of $z^\perp$ through z is a secant of one conic and a passant of the other, or a tangent of both. Equivalently (because every line of $\mathrm{AS}(q)$ of the second kind lies in exactly one such plane, and the fibres through x meet each quadric once), $\mathbf{1}_{\mathcal{E}_1\setminus x} + \mathbf{1}_{\mathcal{E}_2\setminus x}$ is a weighted 2-ovoid of $\mathrm{AS}(q)$: this is Theorem A.*

Proof. A plane through x other than $x^\perp$ meets an elliptic quadric through x with tangent plane $x^\perp$ in a conic through x whose tangent at x is the plane's intersection with $x^\perp$; here that is the line xz, as $x, z \in z^\perp \cap x^\perp$. The lines of $z^\perp$ through z other than xz are the totally isotropic lines through z not through x; the discriminant computation of `POLISHED.md` (proof of Theorem 4.1) shows that the line $\ell(a_0, w_0; d)$ meets $\mathcal{E}_\kappa$ in $1 + \chi(\kappa)\chi\big(N(d)(a_0 - a^*)\big)$ points with the same a^* for both quadrics and $\chi(\kappa_1) = -\chi(\kappa_2)$; this is the secant/passant swap. □

Computer. Squad card `c5` checked (a) both sections are conics through x, (b) xz is tangent to both, (c) the count 2 and (d) the secant/passant/tangent pattern, for every z at $q = 3, 5, 7, 11, 13$ (412 pairs), for every non-square ε ($\varepsilon = 1$ fails, as a control), and re-checked the line form of Theorem A by enumerating all totally isotropic lines.

So the weighted 2-ovoid of Theorem A is a family of *complementary conic pairs*, one pair in each plane $z^\perp$, all through x and sharing their tangent there; the clone of a point $p \in \mathcal{E}_i \setminus x$ stands on the $q + 2$ lines through p, and every line of $\mathrm{AS}(q)$ gains two clones because each line through z is a secant of exactly one conic of the pair, or a tangent of both.

2.4. **The polarity picture: $W(2^{2e+1})$.**

Proposition 2.2. [restatement; proof] *Let $q = 2^{2e+1}$, θ a polarity of $W(q)$, $\mathcal{O}$ its absolute points (a Suzuki–Tits ovoid) and Σ its absolute lines (a spread). The structure $W(q)[\, j\mathbf{1}_{\mathcal{O}},\ j\Sigma\,]$ of Theorem B (j clones of every point of $\mathcal{O}$, j shadow lines over every line of Σ, stand-ins original) is self-polar, with the i-th clone of $p \in \mathcal{O}$ exchanged with the i-th shadow line over $\theta(p) \in \Sigma$. Its polarity graph has diameter at most 3, maximum degree $q + 1 + j$ and $(q + 1 + j)(q^2 + 1)$ vertices.*

Proof. A flag (clone of p, L) with $L \ni p$ goes to $(\theta(L), \text{shadow over } \theta(p))$, and $\theta(L)$ lies on $\theta(p)$; dually for shadow lines. Theorem B gives diameter 4 and Lemma 0.1 diameter 3. □

At $q = 8$ this gives $585, 650, 715$ for $j = 0, 1, 2$, which are the tabled values of the general problem at $(9, 3), (10, 3), (11, 3)$ (Wikipedia table, read 29 October 2026); we have not checked the provenance of those graphs, and expect them to be of this kind. *Computer* (squad card `conics-r2/c2`): for $q = 2$, all 36 polarities of $W(2)$ were enumerated with `nauty`, their absolute points form an ovoid and absolute lines a spread, and for $j = 1, 2$ the structure is self-polar of diameter 4 and folds to 20 and 25 vertices of diameter 3; for $q = 8$ a polarity of $W(8)$ was found in the automorphism group (order 6,340,239,360) and the folds for $j = 1, 2$ have 650 and 715 vertices, maximum degree 10 and 11, diameter 3. For $j = 3$ it gives $780 < 810$. So in quadrangles the polarity picture reproduces known values and gives nothing at the lab's (odd q) cells, where no polarity exists.

3. Generalized hexagons

3.1. **The analogues.** [standard] In a generalized hexagon of order (q,q) with $N=(q+1)(q^4+q^2+1)$ points: a *distance*-2 *ovoid* is a set of q^4+q^2+1 points meeting every line once (pairwise non-collinear; known only for $q=2,3,4$); a *distance*-3 *ovoid* (an *ovoid*) is a set of q^3+1 pairwise opposite points; dually spreads. For the split Cayley hexagon $H(q)\subset Q(6,q)$, two points are opposite iff they are not orthogonal, so partial ovoids of $H(q)$ are partial ovoids of the polar space $Q(6,q)$; ovoids exist for $q=3^h$, not for q even [6] or q prime ≥ 5 [2]. The *Hermitian spread* (the lines of $H(q)$ in a hyperplane meeting $Q(6,q)$ in an elliptic $Q^-(5,q)$) exists for every q [5]. $H(q)$ has a polarity iff $q=3^{2e+1}$; its absolute points form the Ree–Tits ovoid and its absolute lines a spread [7]. The conic-like partial ovoids are the sections $Q^-(3,q)$ (q^2+1 points, every q; their plane sections are conics) and, as found by the lab's squads, unions of conics through one point with a common tangent there (§3.3).

3.2. **Theorem D.** [restatement] Theorem D (`SYNTHETIC.md` §3) doubles $H(q)$ along levels built from a *spread* Σ, one line $v\in\Sigma$ and an exact whisker set. The spread plays the role that the special line plays for the hats: the components of the doubled set are the ball of radius 2 about v and one "broom" per spread line, and the ceiling $q|\mathcal{A}_0|+2(q^3+1)=2N$ is reached because every element outside $\mathcal{A}_0$ sees exactly one element of it. For odd q the whiskers come from q^2-1 further Hermitian spreads through v (with Σ, q^2 spreads pairwise meeting only in v), so the whole configuration is cut out by elliptic hyperplane sections; we found no conic-level description beyond this. The levels are not invariant under a polarity (they contain spread lines but not the corresponding ovoid points), so the folding of §0.2 does not apply to Theorem D as it stands.

A self-polar doubling [new; proof]. For $q=3^{2e+1}$ take the polarity θ, its Ree–Tits ovoid $\mathcal{O}$ and absolute spread Σ, and the single level $\mathcal{A}_0=\mathcal{O}\cup\Sigma$. Its components are the q^3+1 flags $(p,\theta(p))$ (trees of diameter 1), a point off $\mathcal{O}$ is on at most one line of Σ and a line off Σ has at most one point of $\mathcal{O}$, so conditions (L0) and (C) of `SYNTHETIC.md` §3.1 hold and the doubling with m versions has diameter 6, maximum degree $q+m$ and $2N+2(m-1)(q^3+1)$ elements. It is self-polar, so it folds to a graph of diameter 5 with $N+(m-1)(q^3+1)$ vertices: 392 at $(5,5)$ for $q=3$, $m=2$, far below the tabled 650. Not a record; recorded only because it is the hexagon analogue of Corollary 1.7. *Computer* (squad card `conics-r2/c3`): $H(3)$ built from Tits' equations; a polarity found with `nauty` (the automorphism group of the incidence graph has order 8,491,392 $=2|G_2(3)|$); its 28 absolute points are pairwise opposite, its 28 absolute lines pairwise opposite, each absolute point on one absolute line; the doubling has 784 elements, maximum degree 5, diameter 6, is self-polar, and folds to 392 vertices of diameter 5.

3.3. **The diameter-7 freed sets.** [restatement, with one squad computation] The diameter-7 constructions (`SYNTHETIC.md` §4) free the incidences of a matching at a set R of points. With a duality ($q=3^h$) any segment set may be freed, and the point set of a spread (Hermitian, or the Ree–Tits spread when $q=3^{2e+1}$) is the largest one; with arbitrary matchings R must be non-collinear and the matching a local duality, which is easy when R is a partial ovoid. So the size of the records at $(7,7),(9,7),(10,7),(13,7)$ is governed by large partial ovoids of $H(q)$, i.e. of $Q(6,q)$. The conic-like ones are:

- the q^2+1 points of an elliptic section $Q^-(3,q)$ (`D7_HEXAGON.md`, Corollary 9.1, every q);
- *conic fans*: a point e and k conics through e whose planes share one tangent line at e, $1+qk$ points; the points opposite e form an affine space $\mathrm{AG}(5,q)$ in which opposition is $Q'(x-x')\neq 0$ for a quadratic form Q', and each conic is an affine line with non-square Q'-direction (`research/d13D7/REPORT.md` §3, $1+11\cdot 29=320$ at $q=11$);
- the record sets themselves (72, 288, 217 and 756 pairwise opposite points at $q=5,7,8,11$, plus a few mirrored points) come from searches and, at $q=7,11$, from orbits of the Weyl group $W(E_7)$.

Squad computation [computer]: card `c7` rebuilt the conic fans for $q = 5, 7$ (direct check of opposition on all pairs; the $q+1$ points of each fan line with e form a conic of a plane) and searched for large fans with one direction: at $q = 5$ the compatibility graph on the 625 cosets has clique number exactly 7 (exact branch and bound, confirmed by SAT, for every one of the 300 non-square directions), so single-direction fans have at most $1 + 5 \cdot 7 = 36$ points; at $q = 7$ a tabu search found 15 lines, 106 points. The records use 72 and 288 pairwise opposite points at these q, so at small q the conic fans are not competitive; they become useful only at $q = 11$ (320 points), and there the record set (756) is a Weyl-group orbit, not a fan.

What the hexagon offers that the plane does not [heuristic]. In the plane the conic's polarity is what makes the hats self-polar; in the hexagon a polarity exists only for $q = 3^{2e+1}$, outside the lab's cells (which use $q = 5, 7, 8, 9, 11, 13$), and the diameter-7 layouts with an odd number of copies are self-polar only if one copy is mapped to itself, which needs such a polarity. So the conic-as-polarity picture does not reach the hexagon records, and the conic-as-ovoid picture is already what the records use.

4. Summary: what is new

- **New** (proof and computer): the conic frame of the hats (Definition 1.2); the hats in a conic frame are self-polar (Theorem 1.4, Lemma 1.3), and the conic frame is the only frame with this polarity (Remark 1.5; computer for $q = 5, 7$); the gadget read as a change of the conjugacy relation (Proposition 1.6); the explicit regular completion by the polarity (Proposition 1.9); the folded graphs of diameter 2 (Corollary 1.7); the self-polar doubling of $H(3^{2e+1})$ (§3.2).
- **New** (computer): no five or six hats in the hat model at any conic frame for $q \leq 13$; no self-polar five-hat extension of $\mathrm{PG}(2, 5)$ without changes.
- **Restatements**: the dictionary of §0; Theorem A as complementary conic pairs (Proposition 2.1); where ovoids, partial ovoids and spreads sit in Theorems A–C, D and the $D = 7$ freed sets; the $W(8)$ polarity values (Proposition 2.2).
- **Records**: none. No conic-based choice beat a lab value; the folded graphs are below the current general tables.

Appendix A. Files and checks

In `research/synthetic/conics/`: `pg2.py` ($\mathrm{PG}(2, q)$, the conic, the hat rules), `selfpolar.py` and `polar_frames.py` (Theorem 1.4, Corollary 1.7, Proposition 1.9, all polarity types; outputs `*.out`), `sat_hatmodel.py` and `run_hatmodel.py` (hat model at conic frames; `hatmodel_conic.out`), `sat_polar.py` and `run_cases.sh` (self-polar extensions; `k6_q5_E0.out`). Squad round `research/squads/conics-r1/` (brief, eight cards, scoreboard), logged in `research/squads/LOG.md`. Python with `python-sat` (CaDiCaL 1.5.3).

References

[1] G. Araujo-Pardo and N. López, On new record graphs close to bipartite Moore graphs, Graphs Combin. **38** (2022), Paper 110; arXiv:2005.02427.

[2] S. Ball, P. Govaerts, L. Storme, On ovoids of parabolic quadrics, Des. Codes Cryptogr. **38** (2006) 131–145.

[3] M. Ceria, J. De Beule, F. Pavese, V. Smaldore, On large partial ovoids of symplectic and Hermitian polar spaces, J. Combin. Des. **31** (2023) 5–22; arXiv:2203.04553.

[4] S. E. Payne and J. A. Thas, Finite Generalized Quadrangles, 2nd ed., EMS, 2009.

[5] J. A. Thas, Polar spaces, generalized hexagons and perfect codes, J. Combin. Theory Ser. A **29** (1980) 87–93.

[6] J. A. Thas, Ovoids and spreads of finite classical polar spaces, Geom. Dedicata **10** (1981) 135–143.

[7] H. Van Maldeghem, Generalized Polygons, Birkhäuser, 1998.